

For Better or for Health? Medical Knowledge Spillovers and Health Inequality in European Partnerships

Michaela Bunakova¹, Amélie Quesnel-Vallée¹

¹Department of Sociology, McGill University, 885 Sherbrooke Street West, Montreal, Quebec, Canada, H3A 2T7

Corresponding Author: Michaela Bunakova, MSc, PhD(c), Peterson Hall, McGill University, 3460 Mc Tavish St, Montreal, Quebec H3A 0E6. Email: michaela.bunakova@mail.mcgill.ca

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Abstract

Background: Research on health inequalities has largely conceptualized education and socioeconomic resources as individual-level assets. Yet education is embedded within intimate partnerships, where knowledge and competencies may extend beyond the individual and shape the health of close social networks. This paper advances an interpersonal perspective on dyadic health privilege by examining whether, and under what contexts, partnerships with a medical practitioner (MP), representing concentrated health-related expertise, confer health advantages to spouses and cohabiting partners without medical education.

Methods: Using pooled data from 11 rounds of the European Social Survey (2002–2023) across 37 countries ($N = 34,011$), we analyse a restricted sample of socioeconomically advantaged couples to isolate medical knowledge spillovers from general educational homogamy. Mixed-effects linear regression models with random country intercepts are estimated with inverse probability weighting (ATT) to balance treatment and control groups on key sociodemographic characteristics. Interaction models test whether the association is conditioned by macro-level institutional structures, specifically welfare regime type and country-level GDP per capita.

Results: Individuals partnered with medical practitioners report modest but statistically significant improvements in subjective health ($\beta \approx 0.069$ – 0.074 , $p < 0.01$ across adjusted models), corresponding to approximately a 2% increase relative to the sample mean within our 'elite' sample of individuals coupled with highly-educated managers/professionals. Interaction analyses reveal that the health advantage of having an MP partner is most pronounced in lower-income European countries and attenuates significantly, effectively disappearing, in wealthier country contexts, consistent with a macro-level substitution mechanism whereby more comprehensive formal healthcare systems may reduce the marginal value of informal household medical expertise. The association does not vary significantly across welfare regime types, though the Mediterranean and Eastern-European regimes show notable differences.

Conclusions: Medical expertise within partnerships may function as a form of informal health capital, supplementing formal healthcare systems. However, the modest association magnitude and its attenuation at higher levels of national wealth indicate that relational resources are conditioned by broader structural and institutional contexts. These findings highlight the need for universal investment in public health information and navigational support, particularly in lower-income settings where informal household expertise plays a comparatively larger compensatory role.

Keywords: *Health inequality; education spillovers; relational resources; dyads; welfare regimes; medical practitioners; European Social Survey*

1 Introduction

Health inequalities remain a central concern across demographic and public health research, with persistent disparities by education, income, and social position (Deaton, 2002; Mackenbach, 2008; Marmot et al., 1991). Existing scholarship has predominantly conceptualized health-relevant resources, such as educational attainment and occupational status, as individual-level assets. Yet health outcomes are also shaped by resources embedded within close social relationships, particularly intimate partnerships, where knowledge, care, and support are

routinely exchanged (Halpern-Manners et al., 2022a). This relational dimension is increasingly consequential in aging societies where chronic disease management and informal caregiving play growing roles in everyday health maintenance.

This paper examines whether having a spouse or cohabiting partner who is a medical practitioner (MP) is associated with better subjective health. Medical practitioners possess specialized knowledge that may spill over into the household through prevention-oriented advice, early symptom recognition, improved navigation of health systems, and superior informal care. We ask whether individuals partnered with medical practitioners report better subjective health than otherwise compa-

rable individuals partnered with other highly educated professionals, and whether this association is conditioned by macro-level institutional structures, specifically welfare regime type and country-level GDP per capita.

The question matters for at least three reasons. First, it extends health inequality research beyond the individual toward the household as a site of resource transmission (De Neve & Kawachi, 2017). Second, it contributes to work on network health spillovers by focusing on specialized medical knowledge within dyads rather than general educational attainment of one individual alone (Chen et al., 2022). Third, it speaks to political epidemiological concerns with how interpersonal resources interact with institutional contexts to produce unequal health outcomes (Bambra, 2007; Eikemo et al., 2008). Although our empirical focus is deliberately narrow by virtue of being restricted to socioeconomically advantaged couples to minimize confounding from partner education and class, the mechanisms illuminated have implications for understanding how specialized knowledge circulates as a social determinant of health beyond the ‘elite’ circles studied in this paper.

2 Background and theoretical framework

Education as a relational resource

Most empirical applications of the Fundamental Cause framework (Link & Phelan, 1995) treat education as an attribute of individuals. Yet a growing body of evidence reconceives education as an inter-individual good; a shared resource shaped by network members that affects health outcomes of individuals and their families at the same time (Halpern-Manners et al., 2022a). The epidemiological transition toward chronic illness has shifted much health maintenance from hospitals to households, where informal knowledge about diet, prevention, and treatment adherence increasingly determines outcomes (Saracci, 2010). Research documents intergenerational educational spillovers (Friedman & Mare, 2014; Wei et al., 2022) but as a recent comprehensive systematic review suggests, horizontal spillovers within partnerships remain under-explored (De Neve & Kawachi, 2017).

Evidence on spousal education and health consistently finds positive associations. Brown et al. (2014) show that health benefits of own and spousal education combine within marital dyads. More recently, Halpern-Manners et al. (2022) find spousal education is linked to self-rated health, to a greater degree for women than men. Skalická & Kunst (2008) demonstrate that wife’s education predicts men’s all-cause mortality, even after adjusting for the wife’s own socioeconomic characteristics. Critically, however, existing studies focus on highest educational attainment as the crossover exposure, leaving health spillovers of specific types of knowledge largely unexplored (De Neve & Kawachi, 2017).

A closely related strand of research focuses on medical knowledge specifically. Chen et al. (2022) analyse Swedish individuals with a medical professional in their extended family and find lower disease incidence, higher vaccination uptake, and lower smoking rates, consistent with a mechanism of superior health input provision. Our paper builds on this work by

focusing on intimate partnerships rather than extended family ties, and by embedding the analysis in a European comparative framework that allows explicit tests of institutional conditionality.

Macro-level context: welfare regimes and GDP

Institutions shape the health returns to individual and relational resources. The political epidemiology literature demonstrates that welfare regime type is a meaningful predictor of self-reported health (Bambra, 2007; Eikemo et al., 2008) and that macro-level structures condition how effectively individual resources translate into health advantages (Bergqvist et al., 2013). Drawing on Esping-Andersen’s (1990) typology, extended to include Mediterranean (Martin, 1996, 2015) and Eastern European (Beblavy, 2008a) regimes, we expect the health spillover from an MP partner to a non-medically educated partner to be most pronounced in contexts where formal state provision is weakest and families are expected to provide welfare support.

Country-level economic development, as proxied by GDP per capita, provides a complementary macro-level lens. Wealthier countries can invest more in universal healthcare, preventive services, and health education, potentially reducing reliance on informal household expertise (Huijts et al., 2010; Mackenbach, 2008). We therefore hypothesize a substitution mechanism: in lower-income countries where formal healthcare access is more constrained, the marginal value of a medically trained partner to another partner’s health is larger. Conversely, in wealthier countries with comprehensive formal systems, the health advantage of spousal medical expertise may attenuate, consistent with the core logic of Fundamental Cause Theory, which predicts that flexible resources generate advantages particularly where institutional alternatives are scarce (Link & Phelan, 1995; Phelan et al., 2010).

3 Data and Methods

Data source and analytical sample

Data are drawn from eleven rounds of the European Social Survey (ESS; 2002–2023), an academically driven repeat cross-sectional survey administered via face-to-face interviews across European and neighbouring countries (European Social Survey, 2024). The ESS employs probability-based sampling with a target response rate of 70% (Lynn, 2022). All analyses apply the recommended combined ESS design and post-stratification weight, combined with propensity score weights described below.

The analytical sample is restricted to working-age (20–69) respondents who were married or cohabiting and employed at the time of interview, whose partner held a post-secondary qualification or higher and was employed in a managerial or professional occupation based on the highest occupational class category from the International Standard Classification of Occupations, ISCO-08 (ILOSTAT, 2026). Respondents who were themselves medical practitioners were excluded. The treatment group ($N = 866$, 2.5%) comprises respondents partnered with medical practitioners (MPs), including practicing medi-

cal doctors, general practitioners, or specialists. The control group (N = 33,145, 97.5%) comprises respondents partnered with other highly educated managers and professionals. The final analytical sample comprises N = 34,011 individuals across 37 countries.

This design choice, restricting the control group to elite occupational partners only, is deliberate. It minimises confounding by partner educational attainment and occupational prestige, enabling a cleaner identification of medical knowledge as the mechanism of interest rather than general human capital. Several sociodemographic differences between groups remain (Table A1, Appendix A): treatment respondents are more likely to be male (53% vs. 47%), hold post-secondary education (85% vs. 76%), and belong to the highest income quintile (56% vs. 43%), underscoring the need for further propensity score adjustment.

Variables

Outcome. Subjective general health, an established, reliable indicator of morbidity and mortality by decades of social epidemiological research (Ferraro & Farmer, 1999; Idler & Benyamini, 1997), is measured on a five-point Likert scale (1 = very bad, 5 = very good) as part of every ESS survey round collection. The distribution is approximately normal and treated as continuous in this study, following standard practice in cross-national health research.

Treatment. A binary indicator coded 1 if the respondent's partner is a practicing medical practitioner (doctor, GP, or specialist) at the time of interview, 0 otherwise.

Covariates. Individual-level covariates include age category (10-year bands, 40–49 as reference), gender, own and mother's educational attainment, ISCO08 occupational class (five categories, managers/professionals as reference), household income quintile (highest quintile as reference), household size, and respondent's and partner's weekly working hours. Partner education is not included among covariates, given the sample restriction to respondents with highly educated partners only. Macro-level covariates include welfare regime classification (five categories: Social Democratic, Conservative, Liberal, Mediterranean, Post-Communist, the latter as reference) and log GDP per capita from World Bank National Accounts Data (World Bank, n.d.) categorised into three tertiles. Table A2 (Appendix A) provides country-specific details of our welfare regime classificatory framework.

Propensity score weighting

To address non-random selection into treatment, we estimate propensity scores predicting partnership with an MP from respondent age category, gender, and educational and occupational concordance between partners. We construct inverse probability weights in an Average Treatment Effect on the Treated (ATT) configuration: treated individuals receive weight 1; controls are weighted by $p/(1 - p)$. This focuses estimation on the effect of MP partnership among those who actually have an MP partner, and creates a synthetic comparison group resembling the treated in observed covariates. Combined ESS survey and propensity score weights are applied throughout;

this doubly-robust specification re-introduces PS covariates into all regression models.

Statistical models

We estimate six mixed-effects linear regression models with random country intercepts (37 countries), using REML with robust standard errors clustered at the country level. Models 1–4 estimate additive (main effect) specifications progressively adding individual-level controls, welfare regime fixed effects, and GDP tertile. Models 5–6 introduce interaction terms between MP partner status and macro-level institutional structures: welfare regime type (Model 5) and GDP per capita tertile (Model 6). This focus on macro-level moderation is theoretically motivated by the substitution hypothesis: we test whether the health returns to spousal medical expertise are conditioned by the institutional capacity of formal healthcare systems, rather than by individual-level characteristics that are already adjusted for in the additive models. Full model results are presented in Tables 2 and 3, embedded in the relevant results subsections below.

4 Results

Predicted health profiles: own education and medical partner status

To contextualise the magnitude of the medical practitioner partner association, Figure 1 presents predicted subjective health scores for four theoretical analytic profiles derived from our fully adjusted Model 1 estimates. All four profiles represent the same hypothetical respondent, a male, aged 40 to 49, employed as a manager or professional, residing in a two-person household in the highest household income quintile, and working full-time, who is, in every case, partnered with a highly educated manager/professional. The only dimensions that vary across profiles are the respondent's own educational attainment (below versus at or above post-secondary level) and whether the partner holds a medical qualification. This design follows the principle of theoretically anchored marginal prediction (King et al., 2000; Mood, 2010), in which predicted values are computed as the sum of the model intercept and the relevant coefficient(s), holding all other covariates at their reference values.

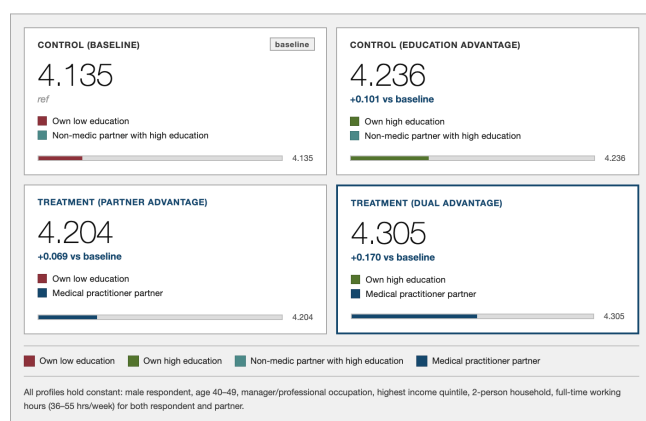


Figure 1. Predicted subjective health scores by education and medical practitioner partner profile. Predicted scores computed as intercept + $\beta_{covariate}$ from Model 1 (fully adjusted mixed-effects linear regression with random intercepts at the country level; $N = 34,011$ across 37 countries; robust standard errors adjusted for clustering at the country level). Intercept = 4.135 (SE = 0.069). Education coefficient: $\beta = +0.101$ (SE = 0.034, $p = 0.003$). MP partner coefficient: $\beta = +0.069$ (SE = 0.025, $p = 0.006$). Design, population, and propensity score weights applied. Data: European Social Survey, pooled rounds 2002–2023.

The baseline profile, a less-educated respondent partnered with a non-medical professional, yields a predicted subjective health score of 4.135 on the five-point scale. This represents the modal “reference person” in the analytical sample: socioeconomically advantaged in terms of partnership and occupational position, but without own post-secondary credentials or proximity to medical expertise in the household. Within the same non-medical partner condition, acquiring post-secondary education or higher raises the predicted score to 4.236, a gain of approximately 2.5 percentage points relative to the maximum possible score of 5. In practical terms, within a population already concentrated at the upper end of the health distribution, this is a meaningful gradient, consistent with the well-documented positive association between own educational attainment and health (Cutler et al., 2006; Mackenbach, 2008).

Substituting an MP partner for a non-medical professional partner, while holding own education fixed at below post-secondary, raises the predicted score from 4.135 to 4.204, a gain corresponding to roughly 68% of the own higher educational premium. This suggests that, for this profile, partnering with a medical practitioner could partially compensate for the absence of own post-secondary attainment as a pathway to better self-reported health. Put differently, a less-educated respondent with an MP partner (predicted score: 4.204) closely approaches the health level of a post-secondary or more educated respondent without a medical partner (4.236). Both groups share the same elite partner profile in terms of occupation and education; what distinguishes them is the source of health-relevant knowledge in the household: formal own higher credentials in one case, and relational proximity to medical expertise in the other. This pattern is consistent with a substitution logic informed by Fundamental Cause Theory (Link & Phelan, 1995; Phelan et al., 2010), whereby flexible health resources, whether acquired through own education or

embedded in an intimate partnership, can generate comparable proximate advantages even when they operate through distinct mechanisms.

The profile combining both own post-secondary education and an MP partner yields the highest predicted score in this comparison: 4.305, representing a cumulative advantage of approximately 4.1% relative to the sample mean, and covers roughly 34% of the entire distance between the sample mean (4.12) and the maximum possible score (5.00).

Taken together, these four profiles illustrate that even within an analytically homogeneous, socioeconomically advantaged subpopulation, one in which every respondent is partnered with a highly educated professional, meaningful variation in predicted subjective health can be traceable to both own educational attainment and the specialised medical knowledge resources accessible through intimate partnerships. The results reinforce the argument that education and knowledge operate not only as individual-level assets but also as interpersonal resources embedded in social networks (De Neve & Kawachi, 2017; Halpern-Manners et al., 2022b) and that their health-protective effects may partially substitute for one another within the household, particularly in contexts of high baseline socioeconomic advantage.

Additive models: predicted health scores across specifications

Across all model specifications, having an MP partner is positively associated with subjective health (see Table 2). In the fully adjusted individual-level model (Model 1; $\beta = 0.069$, $p < 0.01$), the predicted health score for respondents with an MP partner is 4.204, compared to the reference baseline of 4.135. This corresponds to approximately a 1.7% increase relative to the sample mean (4.12). The association remains stable after adding welfare regime fixed effects (Model 2; $\beta = 0.074$) and GDP tertile (Model 3; $\beta = 0.070$), indicating that country-level economic context does not account for the MP partner–health association in the additive specification.

The MP partner effect is modest relative to other structural predictors. The age gradient spans approximately 0.58 points from youngest (Age 20–29: predicted score ≈ 4.49) to oldest (Age 60–69: ≈ 3.91). The income gradient spans approximately 0.33 points between the highest and lowest quintiles. The education premium ($\beta = 0.101$) slightly exceeds the MP partner effect ($\beta = 0.069$), though the gap is small: a respondent with less-than-post-secondary education partnered with a medical practitioner (predicted ≈ 4.204) achieves a health level approaching that of a highly educated respondent without an MP partner (predicted ≈ 4.236). This suggests that relational health capital transmitted through an MP partner can recover approximately two-thirds of the own-education health premium.

Institutional context matters independently of individual characteristics. Relative to Post-Communist regimes, Social Democratic ($\beta = 0.293$) and Liberal regimes ($\beta = 0.364$) are associated with substantially higher subjective health, with gradients substantially larger than the MP partner association. GDP tertile effects are smaller: middle-GDP countries report higher health than the lowest tertile ($\beta = 0.117$, $p < 0.05$),

while the highest tertile coefficient is positive but non-significant ($\beta = 0.100$), suggesting health gains from national wealth may plateau at higher levels of GDP.

Table 2. Multilevel Linear Regression Estimates of Subjective Health: Additive Models 1–3

	Model 1 (Individual)	Model 2 (+Welfare regime)	Model 3 (+GDP tertile)
Medical practitioner partner (ref: no)			
Yes	0.069**	0.074**	0.070**
Age, years (ref: 40–49)			
20–29	0.358***	0.358***	0.360***
30–39	0.180***	0.181***	0.181***
50–59	–0.144***	–0.143***	–0.144***
60–69	–0.224***	–0.224***	–0.224***
Gender (ref: Male)			
Female	0.019	0.018	0.018
Educational attainment (ref: <post-secondary)			
Post-secondary or more	0.101**	0.099**	0.100**
Mother's educational attainment (ref: <post-secondary)			
Post-secondary or more	0.000	–0.003	0.000
Missing	–0.014	–0.019	–0.015
Occupational class (ref: Managers/Professionals)			
Technicians/Clerical	–0.055	–0.055	–0.055
Service/Sales	–0.108	–0.110	–0.110
Agricultural/Crafts/Operators	–0.165**	–0.167**	–0.167**
Elementary Occupations	–0.004	–0.012	–0.006
Household income (ref: Highest quintile)			
Lowest income	–0.330	–0.332	–0.334*
Low-middle income	–0.131*	–0.126*	–0.128**
Middle income	–0.203**	–0.198**	–0.200**
Upper-middle income	–0.144***	–0.142***	–0.143***
Missing income value	–0.151**	–0.143**	–0.145**
Welfare regime (ref: Post-Communist)			
Social Democratic	–	0.293**	–
Conservative	–	0.189	–
Liberal	–	0.364***	–
Mediterranean	–	0.265*	–
Log GDP per capita tertile (ref: Q1 – lowest)			
Q2 – middle	–	–	0.117*
Q3 – highest	–	–	0.100
Constant	4.135***	4.015***	4.078***
Level 1 Variance	0.48	0.48	0.48
Level 2 Variance	0.06	0.04	0.05
ICC	0.11	0.08	0.10

Note. Models also adjusted for household size, and respondent's and partner's working hours; none of these coefficients were statistically significant and are omitted for brevity. Random intercepts at the country level. Robust standard errors adjusted for clustering at the country level (37 clusters). Design, population, and propensity score weights applied. $N = 34,011$. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Interaction models: contextual moderation of the MP partner effect

Additionally, we tested the potential influence of macro-level and contextual factors through interaction models. Table 3 presents abridged regression results for Model 4, which interacts MP partner status with welfare regime, and Model 5, which interacts MP partner status with GDP per capita.

When it comes to welfare regime interactions (Model 4), the MP partner health advantage is largest in Post-Communist countries ($\beta = 0.168$, $p < 0.01$), where the predicted score difference between MP and non-MP respondents is approximately 0.17 points. Interaction terms for Social Democratic, Conservative, and Liberal regimes are negative but non-significant. The interaction for Mediterranean regimes is the exception: negative and statistically significant ($\beta = -0.203$, $p < 0.05$), indicating that in Mediterranean contexts, the health advantage associated with an MP partner effectively disappears (net gap ≈ -0.034). The joint Wald omnibus test is non-significant ($p = 0.29$), cautioning against strong conclusions from individual interaction terms.

The more striking contextual pattern emerges in the GDP tertile interaction model (Model 5). In the lowest GDP tertile countries, the MP partner health advantage corresponds to approximately a 4% increase relative to the sample mean ($\beta = 0.156$, $p < 0.01$). Both the middle ($\beta = -0.132$, $p < 0.05$) and highest ($\beta = -0.132$, $p < 0.05$) GDP tertile interactions are negative and significant, indicating that the health advantage associated with an MP partner shrinks substantially and is statistically indistinguishable from zero in wealthier countries. This macro-level substitution pattern is consistent with a formal healthcare substitution logic: in wealthier countries, well-resourced public health systems, universal coverage, and accessible primary care may effectively substitute for the informal medical capital that a partner provides. Conversely, in lower-income contexts where access is more constrained, proximity to medical expertise within the household plays a comparatively larger compensatory role. These findings are consistent with the core logic of Fundamental Cause Theory (Link & Phelan, 1995; Phelan et al., 2010): flexible resources generate health advantages particularly where institutional alternatives are scarce, and their marginal value diminishes as

formal systems expand.

Table 3. Multilevel Linear Regression Estimates of Subjective Health: Interaction Models 4--5

	Model 4 (Welfare × MP)	Model 5 (GDP × MP)
MP partner (main effect, ref. category)		
Yes	0.168**	0.156**
Welfare regime × MP (ref: Post-Communist)		
Social Democratic	−0.140	−
Conservative	−0.106	−
Liberal	−0.111	−
Mediterranean	−0.203*	−
GDP tertile × MP (ref: Q1 – lowest)		
Q2 – middle	−	−0.132*
Q3 – highest	−	−0.132*
Welfare regime main effects		
Social Democratic	0.293**	−
Conservative	0.189	−
Liberal	0.364***	−
Mediterranean	0.265*	−
GDP tertile main effects		
Q2 – middle	−	0.096
Q3 – highest	−	0.078
Constant	3.962***	4.029***
Level 1 Variance	0.48	0.48
Level 2 Variance	0.04	0.06
ICC	0.08	0.09
Wald Omnibus ($p > \chi^2$)	0.29	0.07

Note. All models include full individual-level covariate adjustment: age, gender, education, mother's education, occupational class, household income, household size, own and partner's working hours, and country-time fixed effects. Random intercepts at the country level. Robust standard errors adjusted for clustering at the country level (37 clusters). Design, population, and propensity score weights applied. $N = 34,011$. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

5 Discussion

This study examined whether intimate partnership with a medical practitioner is associated with better subjective health, and whether that association is conditioned by macro-level contextual structures. Using a restricted sample of socioeconomically advantaged couples across 37 European countries and applying ATT propensity score weighting, we find a small but consistent positive association ($\beta \approx 0.07$ in adjusted models) that persists within our elite subsample characterised by high baseline health.

The persistence of this association after adjustment for demographic, socioeconomic characteristics, and the propensity score reweighting, is consistent with the interpretation of spousal medical expertise as a relational resource that may operate through prevention-oriented norms, superior health-system navigation, and responses to health shocks. From a Fundamental Cause Theory (FCT) perspective (Link & Phelan, 1995), these behavioral pathways are best understood as proximate mechanisms through which a more distal relational resource, medical knowledge embedded in the partnership, operates. Because FCT predicts that fundamental causes continually reproduce inequalities by deploying new proximate pathways, the finding that spousal medical knowledge confers advantage even within an already-privileged sample is consistent with the theory's core logic.

The modest size of the association is itself informative. The MP partner effect ($\beta \approx 0.07$) is substantially smaller than the age gradient (≈ 0.58 points), income gradient (≈ 0.33 points), and welfare regime differences (up to 0.36 points). Within this socioeconomically advantaged subsample, the marginal benefit of medical expertise for everyday health may be limited

when baseline resources and capabilities are high. This is consistent with FCT's prediction that flexible resources operate within, and are partially constrained by, material and institutional conditions.

The interaction analyses provide theoretically meaningful evidence about macro-level context conditionality. While heterogeneity across welfare regime types is limited overall, with the notable exception of Mediterranean contexts where the MP partner advantage attenuates, the GDP pattern offers a coherent substitution story at the macro level. In wealthier countries, comprehensive healthcare systems with shorter wait times, more affordable services and better public health infrastructure might substitute for informal household medical expertise. As such, the MP partner advantage effectively disappears in high-GDP countries. These findings suggest that informal health capital matters most precisely where formal institutional substitutes are least available (Link & Phelan, 1995). This macro-level substitution logic is also consistent with the core proposition of FCT: flexible resources generate health advantages particularly where institutional alternatives are scarce, and their marginal value diminishes as formal systems expand and become more universally accessible.

Our findings also speak to the distribution of health privilege. Medical-partner households are disproportionately concentrated among economically advantaged groups, and the MP partner health advantage is most consequential in lower-income countries where formal systems are least able to compensate. This reinforces the case for investing in universal public health information and navigational support: as formal healthcare infrastructure expands, the health disadvantage of lacking informal medical household expertise diminishes. Policies aimed at reducing health inequalities may thus

yield dual returns, improving average health and reducing inequalities arising from unequal access to relational health capital.

Limitations

Despite statistical adjustment and ATT weighting, unobserved selection remains likely. Medical professionals may attract health-conscious partners, producing residual confounding that cross-sectional data cannot address. Longitudinal designs observing individuals pre- and post-partnership formation would be required to test this mechanism. The ESS also does not consistently include race and religion measures across all waves and countries, limiting our ability to control for important dimensions of assortative mating (Halpern-Manners et al., 2022a). Our outcome, self-rated health, is widely validated in literature (Idler & Benyamini, 1997), but may be sensitive to reporting differences; respondents with MP partners may evaluate their health using more clinical benchmarks, potentially inflating or deflating reported health relative to underlying status. Extending analyses to include objective health markers and health behaviors would strengthen causal interpretation. Finally, our design choice to restrict the sample to elite-partner couples improves internal validity but limits external generalisability to the wider European population or to unpartnered and socially isolated individuals who may lack informal health resources altogether.

6 Conclusion

Using pooled European Social Survey data and a restricted sample of socioeconomically advantaged couples, we find that individuals partnered with medical practitioners report modestly better subjective health than comparable individuals partnered with other highly educated professionals. The association corresponds to approximately a 2% increase relative to the sample mean in fully adjusted models and persists across model specifications that integrate broader systemic characteristics by adding welfare regime and GDP per capita. Tests of macro-level contextual moderation reveal limited heterogeneity across welfare regime types, with the notable exception of Eastern-European and Mediterranean contexts. The health advantage is most pronounced in countries with the lowest GDP per capita and attenuates significantly, approaching zero, in wealthier country contexts, consistent with a macro-level substitution mechanism.

These findings advance understanding of health inequality in three ways. First, they demonstrate that health-relevant resources operate not only at the individual level but also within relational contexts, specifically, intimate partnerships. Second, they provide evidence consistent with a macro-level substitution mechanism: as formal healthcare infrastructure develops and national wealth increases, the health returns to informal relational health capital diminish. Third, they show that even among a privileged analytical sample, the conditions under which spousal medical expertise matters are precisely those in which formal institutional alternatives are most scarce, pointing to a systematic interaction between relational resources and macro-level context.

From a policy perspective, the results suggest that investment in universal access to health information, preventive services, and health-system navigation, particularly in lower-income settings, could both improve average population health and reduce inequalities that arise from the unequal distribution of informal expertise across households. Recognising knowledge as an interpersonal asset embedded in social networks strengthens the case for policy designs that democratise access to high-quality health information rather than leaving such access to depend on the occupation of one's partner.

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Appendix A: Supplementary Tables

Table A1. Analytical sample characteristics by treatment group assignment

	Overall ¹	Treatment (MP partners) ¹	Control (other professionals) ¹
N	34,011 (100%)	866 (2.5%)	33,145 (97.5%)
<i>Subjective health (Likert)</i>			
Very bad	108 (0.3%)	3 (0.3%)	105 (0.3%)
Bad	706 (2.1%)	10 (1.2%)	696 (2.1%)
Fair	5,521 (16%)	114 (13%)*	5,407 (16%)*
Good	16,244 (48%)	418 (48%)	15,826 (48%)
Very good	11,432 (34%)	321 (37%)	11,111 (34%)
<i>Subjective health (binary)</i>			
Very bad/bad/fair	6,335 (19%)	127 (15%)*	6,208 (19%)*
Good/very good	27,676 (81%)	739 (85%)*	26,937 (81%)*
<i>Gender</i>			
Male	16,005 (47%)	461 (53%)*	15,544 (47%)*
Female	18,006 (53%)	405 (47%)*	17,601 (53%)*
<i>Age (years)</i>			
20–29	2,551 (7.5%)	59 (6.8%)	2,492 (7.5%)
30–39	9,638 (28%)	227 (26%)	9,411 (28%)
40–49	10,426 (31%)	248 (29%)	10,178 (31%)
50–59	8,051 (24%)	206 (24%)	7,845 (24%)
60–69	3,345 (9.8%)	126 (15%)*	3,219 (9.7%)*
<i>Working hours/week (respondent)</i>			
<15	1,427 (4.2%)	40 (4.6%)	1,387 (4.2%)
15–35	6,792 (20%)	162 (19%)	6,630 (20%)
36–55	21,798 (64%)	565 (65%)	21,233 (64%)
>55	2,428 (7.1%)	63 (7.3%)	2,365 (7.1%)
Missing	1,566 (4.6%)	36 (4.2%)	1,530 (4.6%)
<i>Partner's working hours/week</i>			
<15	735 (2.2%)	16 (1.8%)	719 (2.2%)
15–35	6,300 (19%)	115 (13%)*	6,185 (19%)*
36–55	22,544 (66%)	575 (66%)	21,969 (66%)
>55	2,514 (7.4%)	113 (13%)*	2,401 (7.2%)*
Missing	1,918 (5.6%)	47 (5.4%)	1,871 (5.6%)
<i>Educational attainment</i>			
<post-secondary	8,018 (24%)	131 (15%)*	7,887 (24%)*
≥ post-secondary	25,993 (76%)	735 (85%)*	25,258 (76%)*
<i>Partner's educational attainment</i>			
≥ post-secondary	34,011 (100%)	866 (100%)	33,145 (100%)
<i>Mother's educational attainment</i>			
<post-secondary	23,550 (69%)	471 (54%)*	23,079 (70%)*
≥ post-secondary	9,476 (28%)	299 (35%)*	9,177 (28%)*
Missing	985 (2.9%)	96 (11%)*	889 (2.7%)*
<i>Educational concordance</i>			
Resp. <post-sec, Partner ≥ post-sec	8,018 (24%)	131 (15%)*	7,887 (24%)*
Both ≥ post-secondary	25,993 (76%)	735 (85%)*	25,258 (76%)*
<i>Occupational class</i>			
Managers/Professionals	19,332 (57%)	568 (66%)*	18,764 (57%)*
Technicians/Clerical	8,964 (26%)	207 (24%)	8,757 (26%)
Service/Sales	2,608 (7.7%)	38 (4.4%)*	2,570 (7.8%)*
Agricultural/Crafts/Ops.	2,650 (7.8%)	43 (5.0%)*	2,607 (7.9%)*
Elementary occupations	457 (1.3%)	10 (1.2%)	447 (1.3%)
<i>Partner's occupational class</i>			
Managers/Professionals	34,011 (100%)	866 (100%)	33,145 (100%)
<i>Occupational concordance</i>			
Both Managers/Professionals	19,332 (57%)	568 (66%)*	18,764 (57%)*
Partner M/P, Respondent lower	14,679 (43%)	298 (34%)*	14,381 (43%)*
<i>Household income (quintiles)</i>			
Lowest	389 (1.1%)	9 (1.0%)	380 (1.1%)
Low-middle	1,636 (4.8%)	26 (3.0%)	1,610 (4.9%)
Middle	3,626 (11%)	51 (5.9%)*	3,575 (11%)*
Upper-middle	7,674 (23%)	157 (18%)*	7,517 (23%)*
Highest	14,641 (43%)	482 (56%)*	14,159 (43%)*
Missing	6,045 (18%)	141 (16%)	5,904 (18%)
<i>Household size</i>			
2 members	10,952 (32%)	306 (35%)	10,646 (32%)
3–4	18,276 (54%)	444 (51%)	17,832 (54%)
5+	4,783 (14%)	116 (13%)	4,667 (14%)
<i>Welfare regime</i>			
Social Democratic	7,644 (22%)	186 (21%)	7,458 (23%)
Conservative	9,262 (27%)	264 (30%)	8,998 (27%)

	Overall ¹	Treatment (MP partners) ¹	Control (other professionals) ¹
Liberal	5,314 (16%)	90 (10%)*	5,224 (16%)*
Mediterranean	2,904 (8.5%)	102 (12%)*	2,802 (8.5%)*
Post-Communist	8,887 (26%)	224 (26%)	8,663 (26%)
<i>Log GDP per capita (tertiles)</i>			
Q1 – lowest	11,439 (34%)	285 (33%)	11,154 (34%)
Q2 – middle	11,334 (33%)	259 (30%)*	11,075 (33%)*
Q3 – highest	11,238 (33%)	322 (37%)*	10,916 (33%)*

¹*n* (%). * $p < 0.01$ row-by-row Pearson χ^2 proportion test. Bold values in the source manuscript indicate significant treatment–control differences.

Table A2. Welfare regime specification and expected health gap between partners of MPs and partners of other educated professionals

Welfare Regime	Countries	State	Market	Family	Expected Health Gap
Liberal/Anglo-Saxon	GBR, IRL, ISR	Marginal	Central	Marginal	Medium: market-centred provision forces reliance on private markets and family.
Conservative/Corporatist	AUT, BEL, CHE, DEU, FRA, LUX, NLD	Subsidiary	Marginal	Central	Significant: family central to welfare; only subsidiary state protection.
Social Democratic	DNK, FIN, ISL, NOR, SWE	Central	Marginal	Marginal	Lower: strong universal state provision and favourable income profile.
Mediterranean/Southern	CYP, ESP, GRC, ITA, PRT	Marginal	Subsidiary	Central	Significant: family and market both central; state plays subsidiary role.
Eastern European	ALB, BGR, CZE, EST, HRV, HUN, LTU, LVA, MKD, MNE, POL, SRB, SVK, UKR, XKX	Central	Marginal	Subsidiary	Medium: accessible state services, but stronger cultural expectations of family care.

Sources for classification: Bambra (2007); Beblavy (2008b); Esping-Andersen (1990); Haggard & Kaufman (2009); Martin (1996, 2015).

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